Algorithms 2

Linear Programming

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1. Linear Programming

1.1 What is Linear Programming?

- **Example:**
  - In a power grid, we have n power plants and m cities. We want to decide how much electricity each plant should produce to meet the demand at minimum cost.
  - Let x_i be the amount of electricity produced by plant i.
  - Let b_j be the demand at city j.
  - Denote by $a_{i,j}$ the amount of electricity from plant i that is sent to city j.
  - For each demand node j, the plants contribute enough power to meet the demand:

$$a_{1,j}x_1 + a_{2,j}x_2 + \ldots + a_{n,j}x_n \geq b_j$$

- If each plant also has a cost c_i, then the total cost is $\sum_{i=1}^{n} c_i x_i$.

1.1 What is Linear Programming?

- This example suggests the general linear-programming template:

$$\underbrace{\boxed{\min_{x\in\mathbb{R}^n} c^t x}}_{\text{objective}}$$

$$\underbrace{\boxed{Ax \leq b, \quad x \geq 0}}_{\text{constraints}}$$

- So a linear program chooses the values of $x_1, \ldots, x_n$ to optimize a linear objective while satisfying linear constraints. More generally, the objective can be written as either a minimization or a maximization problem.

1.2 IP's and IP's in Matricial form.

Definition 1 An Integer program is an optimization problem over $\boldsymbol{x} \in \mathbb{Z}^n$ that can be written in one of the following six forms:

$$\begin{array}{lll} \max\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} \le \boldsymbol{b}\} & \max\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} \le \boldsymbol{b}, \boldsymbol{x} \ge \boldsymbol{0}\} & \max\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \ge \boldsymbol{0}\} \\ \min\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} \ge \boldsymbol{b}\} & \min\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} \ge \boldsymbol{b}, \boldsymbol{x} \ge \boldsymbol{0}\} & \min\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \ge \boldsymbol{0}\} \end{array}$$

Definition 2 A linear program is an optimization problem over $\boldsymbol{x} \in \mathbb{R}^n$ that can be written in one of the following six forms:

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Remark

LP's $\in$ P but IP's $\in$ NPH

2. Examples

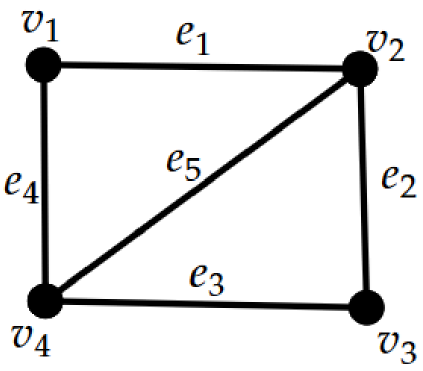
2.1 Incidence matrix

The *incidence* matrix of a graph G is given by

$$\begin{array}{c} \\ v_1 \\ v_2 \\ \vdots \\ v_n \end{array} \begin{array}{c} e_1 \; e_2 \; \dots \; e_m \\ \left[\begin{array}{ccc} & & \\ & 0/1 & \\ & & \\ & & \end{array} \right] \end{array} \qquad M_{ve} := \begin{cases} 1, & v \in e \\ 0, & v \notin e \end{cases}$$

$$M := M(G)$$

$$\begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \end{array} \begin{array}{c} \begin{array}{ccccc} e_1 & e_2 & e_3 & e_4 & e_5 \end{array} \\ \left[\begin{array}{ccccc} 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{array} \right] \end{array}$$



Each column has $\#1 = 2$
Each row has $\#1 = \deg(v)$

2.2 Maximum Matching

Given G with n vertices and m edges, we seek a vector $\boldsymbol{x} \in \{0, 1\}^m$ s.t.

- $x_e = 1$ when edge e belongs to the matching, and $x_e = 0$ otherwise.
- maximize $\sum_{e \in E} x_e = \max \mathbf{1}^t \boldsymbol{x}$. **(We want maximum matching)**

This vector $\boldsymbol{x}$ is the characteristic vector of a matching in G.

So far the above doesn't define a matching!
We need to define constraints.

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or by matricial form

$$\boxed{\begin{array}{c} \boldsymbol{x} \in \mathbb{Z}^n \\ \max \mathbf{1}^t \boldsymbol{x} \end{array}}$$

$$\begin{array}{cc} & \begin{array}{cccccc} e_1 & e_2 & & \cdots & & e_m \end{array} \\ \begin{array}{c} v_1 \\ v_2 \\ \vdots \\ v_n \end{array} & \underset{n \times m}{\underset{M}{\begin{bmatrix} 1 & 1 & 0 & \cdots & 0 & 1 \\ &&&&& \\ &&&&& \\ &&&&& \end{bmatrix}}} \end{array} \underset{m \times 1}{\underset{x}{\begin{bmatrix} 1 \\ 0 \\ 1 \\ \vdots \\ 1 \\ 1 \end{bmatrix}}} = \underset{n \times 1}{\underset{Mx}{\begin{bmatrix} (Mx)_1 \\ \vdots \\ (Mx)_i \\ \vdots \end{bmatrix}}}$$

$(Mx)_i = \langle v_i, x \rangle$

#edges incident to v_i by the given matching

This is why $Mx \leq 1$ must hold if X is a matching!

2.2 Maximum Matching

For the max-matching problem, we arrive at the following

IP for maximum matching

$$\max_{x \in \mathbb{Z}^m} \{\mathbf{1}^t \boldsymbol{x} : M\boldsymbol{x} \leq \mathbf{1}, \boldsymbol{x} \geq \mathbf{0}\}$$

LP for maximum matching

$$\max_{\boldsymbol{x} \in \mathbb{R}^m} \{\mathbf{1}^t \boldsymbol{x} : M\boldsymbol{x} \leq \mathbf{1}, \boldsymbol{x} \geq \mathbf{0}\}$$

This called the:
Corresponding LP Relaxation

Remark

We dont need to bound $\boldsymbol{x} \leq \mathbf{1}$ as if there exists i for which $x_i > 1$. Then, there must exist $j \in [m]$ for which $M_{j,i} < 1$. But, we know that M is the *incidence matrix* of G so that for exactly two indicies j_1, j_2 we have $M_{j_1,i} = 1$ and $M_{j_1,i} = 1$.

2.3 Vertex Cover

Given G with n vertices and m edges, we seek a vector $\boldsymbol{y} \in \{0,1\}^n$ s.t.

- $y_v = 1$ when vertex v belongs to the **vc**, and $y_v = 0$ otherwise.
- minimize $\sum_{v \in V} y_v = \min \mathbf{1}^t \boldsymbol{y}$. **(We want minimum vertex-cover)**

Constaints: $\forall e \in E : \sum_{v \in e} y_v \geq 1$ **(Every edge is covered at least once)**

This vector $\boldsymbol{y}$ is the characteristic vector of a vertex cover in G.

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or by matricial form

$$\boxed{\begin{array}{c} \boldsymbol{y} \in \mathbb{Z}^n \\ \min \mathbf{1}^t \boldsymbol{y} \end{array}}$$

$$\underset{\substack{M^t \\ n \times m}}{\begin{array}{c|c} & \begin{array}{cccccc} v_1 & v_2 & & \cdots & & v_n \end{array} \\ \begin{array}{c} e_1 \\ e_2 \\ \vdots \\ e_m \end{array} & \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 1 \\ & & & & & \\ & & & & & \\ & & & & & \end{bmatrix} \end{array}} \underset{\substack{y \\ n \times 1}}{\begin{bmatrix} 1 \\ 1 \\ 0 \\ \vdots \\ 1 \\ 0 \end{bmatrix}} = \underset{\substack{My \\ m \times 1}}{\begin{bmatrix} (M^t y)_1 \\ \vdots \\ (M^t y)_i \\ \vdots \end{bmatrix}}$$

$(M^t y)_i = \langle e_i, y \rangle$

#vertices covering the edge e_i

This is why $M^t y \geq 1$ must hold if Y is a matching!

For the min-vc problem, we arrive at the following

IP for minimum-vc

$$\min_{\boldsymbol{y}\in\mathbb{Z}^n}\{\mathbf{1}^t\boldsymbol{y} : M^t\boldsymbol{y} \geq \mathbf{1}, \boldsymbol{y} \geq \mathbf{0}\}$$

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$$\underbrace{\min_{\boldsymbol{y}\in\mathbb{R}^n}\{\mathbf{1}^t\boldsymbol{y} : M\boldsymbol{y} \geq 1, \boldsymbol{y} \geq 0\}}_{\substack{\text{This called the:}\\ \textbf{Corresponding LP Relaxation}}}$$

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What if we wanted vc with minimum weight? i.e. every vertex $v \in V$ has weight $w_v \in \mathbb{R}$?

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$$\min_{\boldsymbol{y}\in\mathbb{Z}^n}\left\{\underbrace{\boldsymbol{w}^t\boldsymbol{y}}_{\sum_{v\in V} w_v\cdot y_v} : M^t\boldsymbol{y} \geq \mathbf{1}, \boldsymbol{y} \geq \mathbf{0}\right\}$$

2.4 Vertex Cover vs Maximum Matching

By Konig we know that in bipartite graphs it must hold

IP for maximum matching

$$\max_{\boldsymbol{x} \in \mathbb{Z}^n} \{\mathbf{1}^t \boldsymbol{x} : M\boldsymbol{x} \leq \mathbf{1}, \boldsymbol{x} \geq \mathbf{0}\}$$

$$=$$

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LP for maximum matching

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If G is not bipartite, then in fact

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$$\max_{\boldsymbol{x}\in\mathbb{Z}^n}\{\mathbf{1}^t\boldsymbol{x} : M\boldsymbol{x} \leq \mathbf{1}, \boldsymbol{x} \geq \mathbf{0}\}$$

$$\in \mathrm{P}$$

IP for minimum-vc

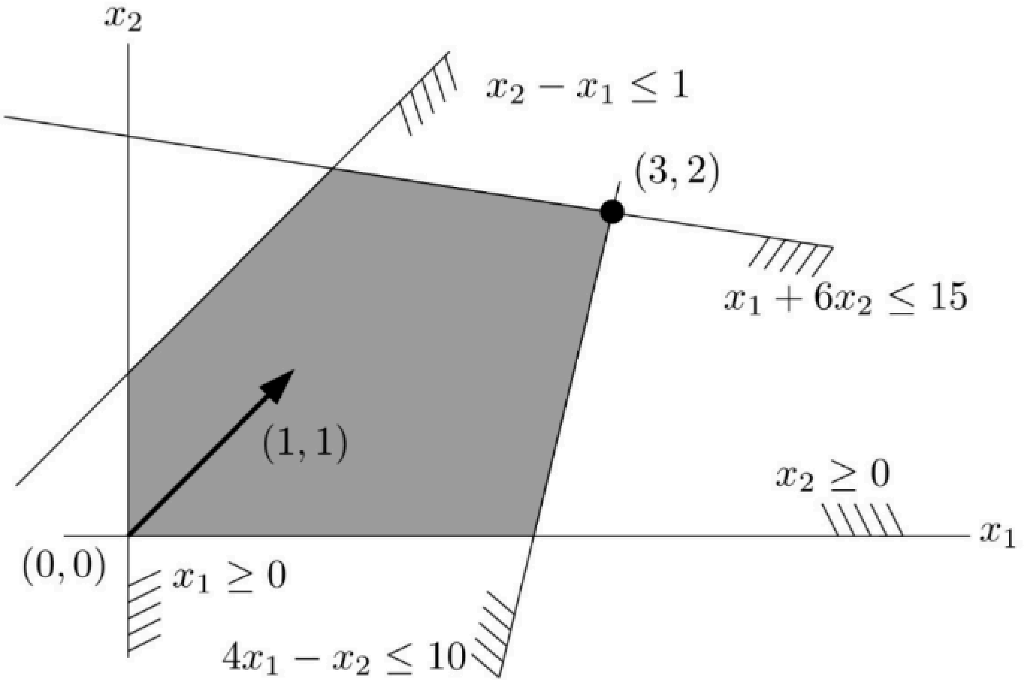
$$\min_{\boldsymbol{y}\in\mathbb{Z}^n}\{\mathbf{1}^t\boldsymbol{y} : M^t\boldsymbol{y} \geq \mathbf{1}, \boldsymbol{y} \geq \mathbf{0}\}$$

$$\in \mathrm{NPC}$$

3. Geometric analysis of LP's

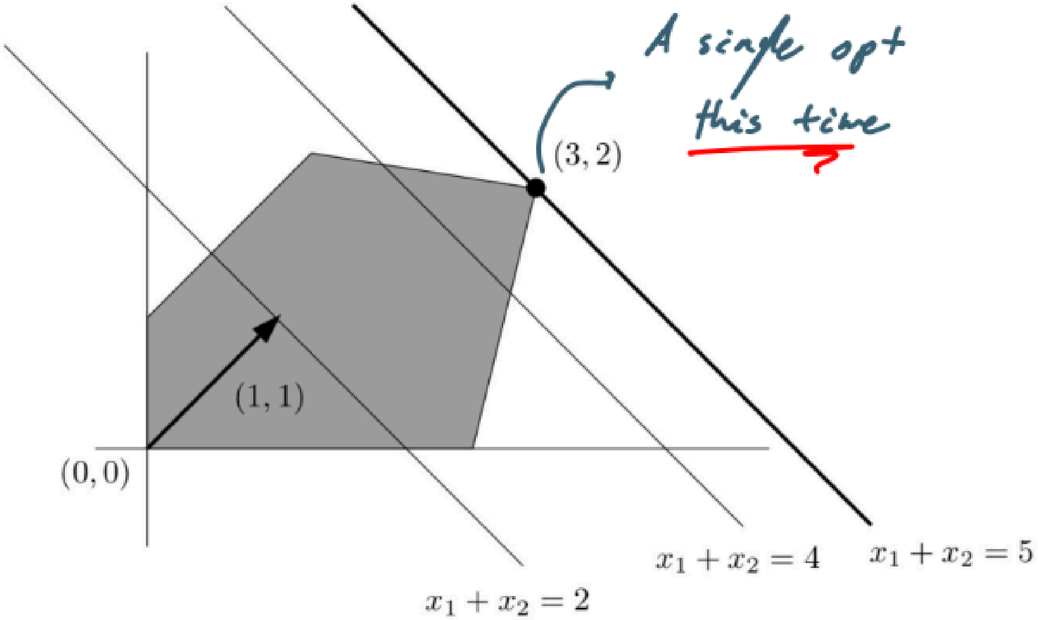
Consider the following **LP**:

$$\max x_1 + x_2$$
$$s.b.t.$$
$$\begin{aligned} -x_1 + x_2 &\leq 1 \\ x_1 + 6x_2 &\leq 15 \\ 4x_1 - x_2 &\leq 10 \\ x_1 &\geq 0 \\ x_2 &\geq 0 \end{aligned}$$



$$\max_{x \in \mathbb{R}^2} (1,1) \cdot (x_1, x_2) \text{ s.t. } \begin{pmatrix} -1 & 1 \\ 4 & 6 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 1 \\ 15 \\ 10 \end{pmatrix}, \quad (x_1, x_2) \geq (0,0)$$

Solving the problem Geometrically, would mean taking the line $x_1 + x_2 = c$ and moving it in the direction $(1, 1)$ until we get out of constraint.



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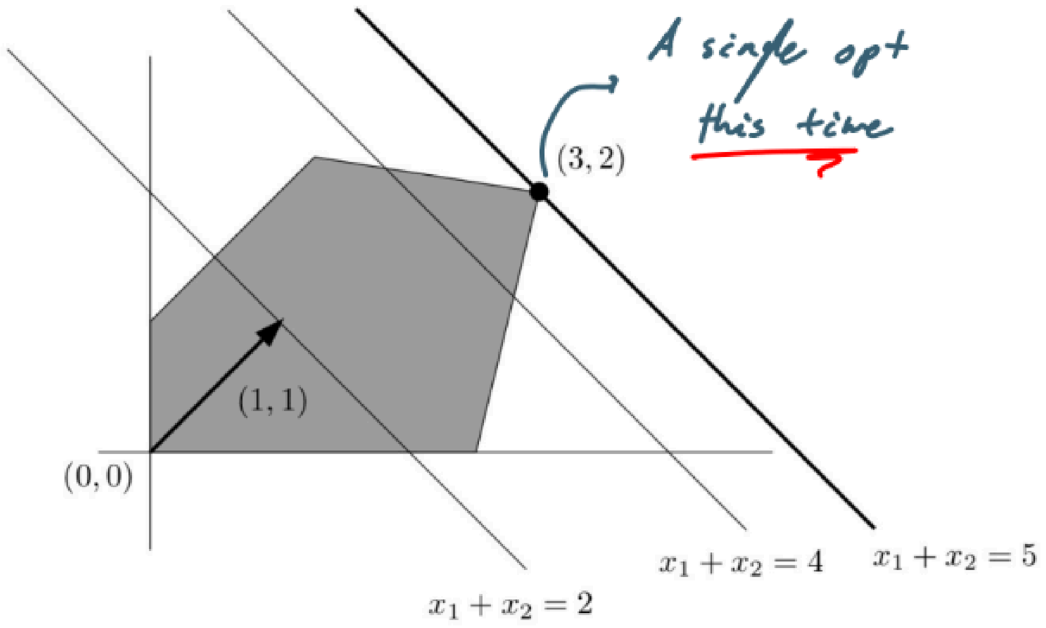

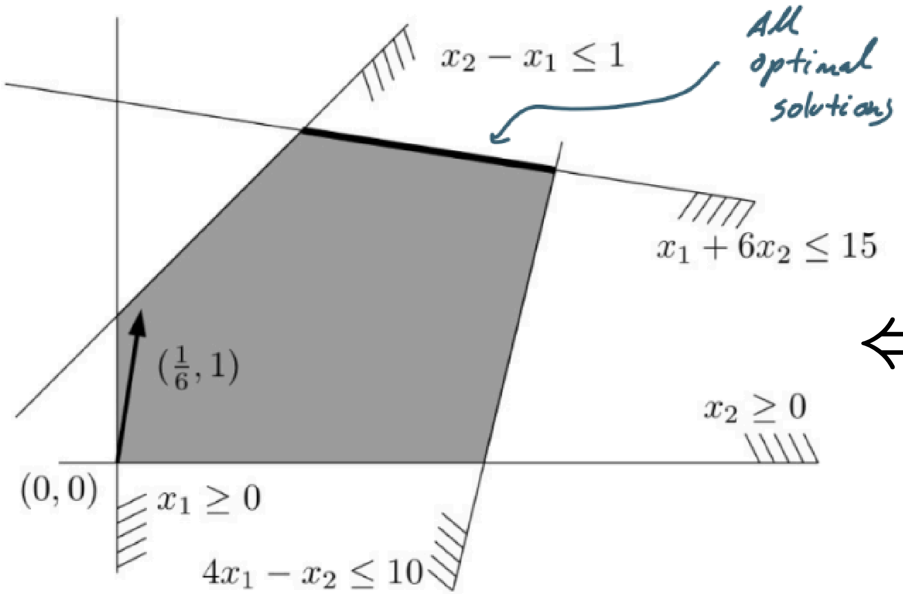

$\Longleftarrow$ If we change the constraint to max $\frac{1}{6}x_1 + x_2$. Then, we have infinitely maxny solutions.

3.2 Feasibility and Boundedness

An LP can fail to have an optimal solution in two ways:

Infeasible — flipping two constraints of our example:

$$\max x_1 + x_2$$
$$x_2 - x_1 \geq 1$$
$$x_1 + 6x_2 \leq 15$$
$$4x_1 - x_2 \geq 10$$
$$x_1, x_2 \geq 0$$

(constraints 1 & 3 flipped)

The constraints have **no common intersection** — the LP is *infeasible*.

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Unbounded — removing two constraints:

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Feasible solutions exist but the objective grows **without bound** — the LP is *unbounded*.

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Theorem 3

Any **feasible** and **bounded** LP has an **optimal** solution.

3.3 Standard forms of LP's

Definition 3 A Linear program in one of the following 2-forms:

$$\max\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \geq \boldsymbol{0}\} \qquad \min\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \geq \boldsymbol{0}\}$$

is said to be in **equational form**;

Remark

Optimal solution for **LP's** in **equational form** are the easiest to exaplain and we will focus on those.

3.4 Converting to Equational Form

Any LP can be converted to equational form. Consider a **mixed** LP (not in any standard form):

$$\max 3x_1 - 2x_2$$

$$\begin{aligned} 2x_1 - x_2 &\leq 4 \\ x_1 + 3x_2 &\geq 5 \\ x_1 &\geq 0 \end{aligned}$$

(x_2 is **free** – no sign constraint)

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Step 1 ($\le$): introduce *slack variable* $x_3 \ge 0$ and set:

$$2x_1 - x_2 \le 4 \Rightarrow 2x_1 - x_2 + x_3 = 4,$$

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Final equational LP over $(x_1, y_1, y_2, x_3, x_4 \ge 0)$:

$$\max 3x_1 - 2y_1 + 2y_2$$

$$\begin{aligned} 2x_1 - y_1 + y_2 + x_3 &= 4 \\ -x_1 - 3y_1 + 3y_2 + x_4 &= -5 \\ x_1, y_1, y_2, x_3, x_4 &\ge 0 \end{aligned}$$

By solving the final LP we can derive the optimal values for the original LP

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Remark

Non-negativity constraints ($x \geq 0$) are called **non-negativity constraints**. The forms with $=$ and $\boldsymbol{x} \geq \mathbf{0}$ are the simplest to analyze.

4. LP’s in Equational Form

4.1 Basic Solutions

We study LPs of the form $\max\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \geq \boldsymbol{0}\}$ where $A \in M_{m\times n}(\mathbb{R})$ with $\operatorname{rank}(A) = m$ (so $m \leq n$).

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Definition 5 $\boldsymbol{x} \in \mathbb{R}^n$ satisfying $A\boldsymbol{x} = \boldsymbol{b}$ is called **basic** if $\exists B \subseteq [n]$ with $|B| = m$ s.t.:

- A_B is **non-singular** (columns of A indexed by B are linearly independent), and
- $x_j = 0 \quad \forall j \notin B$.

A basic $\boldsymbol{x}$ satisfying also $\boldsymbol{x} \geq \boldsymbol{0}$ is a **basic feasible solution** (BFS).

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- A_B is **non-singular** (columns of A indexed by B are linearly independent), and
- $x_j = 0 \quad \forall j \notin B$.

A basic $\boldsymbol{x}$ satisfying also $\boldsymbol{x} \geq \boldsymbol{0}$ is a **basic feasible solution** (BFS).

Remark

Basic solutions may **fail** non-negativity — hence the distinction.

$$\underbrace{\begin{bmatrix} -1 & 0 & 1 & 3 & 0 \\ 4 & 0 & 2 & -1 & 1 \\ 0 & 1 & 2 & 0 & -2 \end{bmatrix}}_{A\in\mathbb{R}^{3\times5}} \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}}_{\vec{x}} = \underbrace{\begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}}_{\vec{b}} \xRightarrow[B=[1,2,4]]{} \underbrace{\begin{bmatrix} -1 & 0 & 3 \\ 4 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}}_{A_B\in\mathbb{R}^{3\times3}} \underbrace{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}}_{\vec{x_B}} = \underbrace{\begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}}_{\vec{b}} \implies \begin{matrix} \vec{x} \text{ is} \\ \text{basic feasible} \end{matrix}$$

Observation 1 Let $\boldsymbol{x} \in \mathbb{R}^n$ be feasible. Set $K := \{j \in [n] : x_j > 0\}$. Then

$$\boldsymbol{x} \text{ is basic} \iff \text{columns of } A_K \text{ are linearly independent.}$$

Proof of observation:

($\Longrightarrow$): $\boldsymbol{x}$ is basic, so $\exists B$ with $|B| = m$, A_B non-singular, and $x_j = 0$ for $j \notin B$. Since $\boldsymbol{x} \geq \mathbf{0}$ we have $K \subseteq B$, so columns of A_K are a subset of the independent columns of A_B – hence independent.

Observation 2 Let $\boldsymbol{x} \in \mathbb{R}^n$ be feasible. Set $K := \{j \in [n] : x_j > 0\}$. Then

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($\impliedby$): Extend the columns of A_K to a maximal independent set B in A. Since $\mathrm{rk}(A) = m$, we get $|B| = m$, so A_B is non-singular. As $K \subseteq B$, we have $x_j = 0$ for all $j \notin B$. □

Algorithm – to decide if a feasible $\boldsymbol{x}$ is basic:

Input: feasible $\boldsymbol{x} \in \mathbb{R}^n$, i.e. $\boldsymbol{x} \geq \mathbf{0}$ and $A\boldsymbol{x} = \boldsymbol{b}$.

1. Set $K = \{j \in [n] : x_j > 0\}$.
2. If columns of A_K are independent, return **basic**; else return **not basic**.

Lemma 4 Any $B \subseteq [n]$ with $|B| = m$ has **at most one** basic feasible solution fitting it.

Proof: Let $N = [n] \setminus B$, and fix a solution $\boldsymbol{x} \in \mathbb{R}^n$ satisfying $A\boldsymbol{x} = \boldsymbol{b}$.

- If $\boldsymbol{x}$ fits $B \implies$ then $\boldsymbol{x}_N = \boldsymbol{0}$.
- so $A_B\boldsymbol{x}_B + A_N\boldsymbol{x}_N = \boldsymbol{b} \implies A_B\boldsymbol{x}_B = \boldsymbol{b}$.
- Since A_B is non-singular, $\boldsymbol{x}_B = A_B^{-1}\boldsymbol{b}$ is uniquely determined. $\square$

4.2 Properties of Basic Feasible Solutions

Proposition 4.2.2 There are at most $\binom{n}{m}$ basic feasible solutions.

Proposition 4.2.4 There are at most $\binom{n}{m}$ basic feasible solutions.

This alone is not useful unless optimal solutions are **basic**:

Theorem 8 Let LP $\max\{\boldsymbol{c}^T\boldsymbol{x} : \boldsymbol{x} \geq \boldsymbol{0}, A\boldsymbol{x} = \boldsymbol{b}\}$ be feasible and bounded.
For every feasible $\boldsymbol{x}_0$, there exists a BFS $\overline{\boldsymbol{x}}$ s.t. $\boldsymbol{c}^T\overline{\boldsymbol{x}} \geq \boldsymbol{c}^T\boldsymbol{x}_0$.

Proposition 4.2.6 There are at most $\binom{n}{m}$ basic feasible solutions.

This alone is not useful unless optimal solutions are **basic**:

Theorem 10 Let LP $\max\{\boldsymbol{c}^T\boldsymbol{x} : \boldsymbol{x} \geq \boldsymbol{0}, A\boldsymbol{x} = \boldsymbol{b}\}$ be feasible and bounded.
For every feasible $\boldsymbol{x}_0$, there exists a BFS $\overline{\boldsymbol{x}}$ s.t. $\boldsymbol{c}^T\overline{\boldsymbol{x}} \geq \boldsymbol{c}^T\boldsymbol{x}_0$.

Applying this to an optimal $\boldsymbol{x}_0$ (which exists since the LP is feasible and bounded):

$$\exists \text{ an optimal basic feasible solution.}$$

We now have an **exponential algorithm**: enumerate all $\binom{n}{m}$ bases, find each of the **Basic Feasible Solutions**, and pick the best.

5. Geometric View

5.1 The Feasible Polyhedron

The linear constraints of a feasible, bounded LP form a **convex polyhedron**:

Convex: any line segment between two feasible points stays inside the region.

Every half-space $\{\boldsymbol{x} : \boldsymbol{a}^T\boldsymbol{x} \leq b\}$ is convex. The feasible region — as their intersection — is convex too.

Not-convex: a line segment between two feasible points may **exit** the region.

The feasible region of an **IP** is generally not convex (integer lattice points only).

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- The proposition above guarantees an optimal solution at a **vertex**.

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- **Basic feasible solutions** lie on the **vertices** and **edges** of the polyhedron.
- The proposition above guarantees an optimal solution at a **vertex**.

Two classical algorithms traverse the vertices in smarter ways:

Simplex Algorithm

Moves along edges of the polyhedron, improving the objective at each step. Exponential worst-case, but very fast in practice.

Ellipsoid Algorithm

Approximates the feasible region with shrinking ellipsoids. First provably polynomial-time LP algorithm.

6. Duality

6.1 LPs Come in Pairs

Every LP (the **primal**) has a companion LP called its **dual**.

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We have already seen duality:

König's Theorem: $\tau(G) = \nu(G)$ in bipartite graphs

$$\text{max matching} = \text{min vertex cover}$$

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Why is duality useful?

- It provides **certificates of optimality** for feasible primal solutions.
- It enables LP-based algorithms **without** invoking expensive methods like Simplex or Ellipsoid.
  - Dijkstra's algorithm was designed exactly this way.

Example: Consider the LP

$$
\begin{aligned}
&\max 5x_1 + 5x_2 \\
&s.b.t. \quad & 3x_1 + x_2 &\leq 7 \\
& & 2x_1 + 4x_2 &\leq 8 \\
& & x_1, x_2 &\geq 0
\end{aligned}
$$

Geometrically, $(1, 2)$ is optimal with objective value $5 \cdot 1 + 5 \cdot 2 = 15$.

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$$\begin{aligned} &\max 5x_1 + 5x_2 \\ &s.b.t. \quad & 3x_1 + x_2 \leq 7 \\ && 2x_1 + 4x_2 \leq 8 \\ && x_1, x_2 \geq 0 \end{aligned}$$

Geometrically, $(1, 2)$ is optimal with objective value $5 \cdot 1 + 5 \cdot 2 = 15$.

Q: How do we **prove** $(1, 2)$ is optimal without drawing a picture?

A: Sum the two constraints (both multiplied by 1):

$$(3x_1 + x_2 \leq 7) + (2x_1 + 4x_2 \leq 8) \quad \Rightarrow \quad 5x_1 + 5x_2 \leq 15.$$

Every feasible solution satisfies $5x_1 + 5x_2 \leq 15$, and $(1, 2)$ achieves 15 – so it is **optimal**. $\square$

6.1 LPs Come in Pairs

Now change the objective to max $7x_1 + 4x_2$ (same constraints):

$$
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&\max 7x_1 + 4x_2 \\
&s.b.t. \qquad & 3x_1 + x_2 &\leq 7 \\
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A: Look for weights $y_1, y_2 \geq 0$ such that the weighted sum of constraints **matches** the objective:

$$\begin{aligned} & y_1(3x_1 + x_2 \leq 7) + y_2(2x_1 + 4x_2 \leq 8) \\ \Rightarrow \quad & \underbrace{(3y_1 + 2y_2)}_{=7}x_1 + \underbrace{(y_1 + 4y_2)}_{=4}x_2 \leq 7y_1 + 8y_2 \end{aligned}$$

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Solving $3y_1 + 2y_2 = 7$ and $y_1 + 4y_2 = 4$ gives $y_1 = 2$, $y_2 = \frac{1}{2}$.

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\Rightarrow \quad & \underbrace{(3y_1 + 2y_2)}_{=7} x_1 + \underbrace{(y_1 + 4y_2)}_{=4} x_2 \leq 7y_1 + 8y_2
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Solving $3y_1 + 2y_2 = 7$ and $y_1 + 4y_2 = 4$ gives $y_1 = 2$, $y_2 = \frac{1}{2}$.

Upper bound: $7y_1 + 8y_2 = 7 \cdot 2 + 8 \cdot \frac{1}{2} = 14 + 4 = 18$.

$(2,1)$ is feasible ($3 \cdot 2 + 1 = 7$, $2 \cdot 2 + 4 \cdot 1 = 8$) and achieves $7 \cdot 2 + 4 \cdot 1 = 18$ – so it is **optimal**. □

6.2 Constructing the Dual

Given the LP $\max\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} \leq \boldsymbol{b}\}$ (m constraints, n variables):

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1. Assign a **dual variable** $y_i \geq 0$ to each constraint $\boldsymbol{a}_i^T\boldsymbol{x} \leq b_i$.
2. Multiply constraint i by y_i and **sum**:

$$\sum_i y_i(\boldsymbol{a}_i^T\boldsymbol{x}) \leq \sum_i y_i b_i \quad \Rightarrow \quad \left(A^T\boldsymbol{y}\right)^T\boldsymbol{x} \leq \boldsymbol{y}^T\boldsymbol{b}.$$

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Observation 7 $A \in M_{m\times n}(\mathbb{R})$, $\boldsymbol{c} \in \mathbb{R}^n$, $\boldsymbol{b} \in \mathbb{R}^m$. If $A^T\boldsymbol{y} = \boldsymbol{c}$ and $\boldsymbol{y} \ge \boldsymbol{0}$, then $\boldsymbol{c}^T\boldsymbol{x} \le \boldsymbol{y}^T\boldsymbol{b}$ whenever $A\boldsymbol{x} \le \boldsymbol{b}$.

Proof: $\boldsymbol{y}^T\boldsymbol{b} \ge \boldsymbol{y}^T(A\boldsymbol{x}) = \left(A^T\boldsymbol{y}\right)^T\boldsymbol{x} = \boldsymbol{c}^T\boldsymbol{x}$. □

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3. To obtain an upper bound on $\boldsymbol{c}^T\boldsymbol{x}$, require $A^T\boldsymbol{y} = \boldsymbol{c}$; then $\boldsymbol{c}^T\boldsymbol{x} \leq \boldsymbol{y}^T\boldsymbol{b}$.

Observation 8 $A \in M_{m\times n}(\mathbb{R})$, $\boldsymbol{c} \in \mathbb{R}^n$, $\boldsymbol{b} \in \mathbb{R}^m$. If $A^T\boldsymbol{y} = \boldsymbol{c}$ and $\boldsymbol{y} \geq \boldsymbol{0}$, then $\boldsymbol{c}^T\boldsymbol{x} \leq \boldsymbol{y}^T\boldsymbol{b}$ whenever $A\boldsymbol{x} \leq \boldsymbol{b}$.

Proof: $\boldsymbol{y}^T\boldsymbol{b} \geq \boldsymbol{y}^T(A\boldsymbol{x}) = (A^T\boldsymbol{y})^T\boldsymbol{x} = \boldsymbol{c}^T\boldsymbol{x}$. □

Theorem 16 Weak Duality Theorem

$$\max\{\boldsymbol{c}^T\boldsymbol{x} : A\boldsymbol{x} \leq \boldsymbol{b}\} \leq \min\{\boldsymbol{y}^T\boldsymbol{b} : A^T\boldsymbol{y} = \boldsymbol{c}, \boldsymbol{y} \geq \boldsymbol{0}\}$$

whenever both LPs are feasible and bounded.

6.2 Constructing the Dual

The Primal → Dual recipe: each primal **constraint** → dual **variable**; each primal **variable** → dual **constraint**.

Primal constraint → dual variable

$$
\begin{aligned}
\langle \boldsymbol{a}_i, \boldsymbol{x} \rangle \le b_i &\quad \Rightarrow y_i \ge 0 \\
\langle \boldsymbol{a}_i, \boldsymbol{x} \rangle = b_i &\quad \Rightarrow y_i \in \mathbb{R} \\
\langle \boldsymbol{a}_i, \boldsymbol{x} \rangle \ge b_i &\quad \Rightarrow y_i \le 0
\end{aligned}
$$

Primal variable → dual constraint

$$
\begin{aligned}
x_i \ge 0 &\quad \Rightarrow \left(A^T \boldsymbol{y}\right)_i \ge c_i \\
x_i \le 0 &\quad \Rightarrow \left(A^T \boldsymbol{y}\right)_i \le c_i \\
x_i \in \mathbb{R} &\quad \Rightarrow \left(A^T \boldsymbol{y}\right)_i = c_i
\end{aligned}
$$

Applying the recipe to our LP:

(P)

$$\boxed{\begin{aligned} \max 5x_1 + 5x_2 \\ 3x_1 + x_2 \le 7 \\ 2x_1 + 4x_2 \le 8 \\ x_1, x_2 \ge 0 \end{aligned}} \leftrightarrow \boxed{\begin{aligned} &(D) \quad \min 7y_1 + 8y_2 \\ &3y_1 + 2y_2 \ge 5 \\ &y_1 + 4y_2 \ge 5 \\ &y_1, y_2 \ge 0 \end{aligned}}$$

6.3 Examples

Applying the recipe to our LP:

$$(P) \quad \boxed{\begin{aligned} \max 5x_1 + 5x_2 \\ 3x_1 + x_2 \le 7 \\ 2x_1 + 4x_2 \le 8 \\ x_1, x_2 \ge 0 \end{aligned}} \leftrightarrow \boxed{\begin{aligned} &(D) \quad \min 7y_1 + 8y_2 \\ &3y_1 + 2y_2 \ge 5 \\ &y_1 + 4y_2 \ge 5 \\ &y_1, y_2 \ge 0 \end{aligned}}$$

Remark

The dual of the dual is the primal.

6.4 Strong Duality

Weak duality gives an upper bound. The true connection is **much stronger**:

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Weak duality gives an upper bound. The true connection is **much stronger**:

Theorem 18 Strong Duality Theorem

Let (P) and (D) be a primal LP and its dual. **Precisely one** of the following holds:

1. Both (P) and (D) are **infeasible**.
2. (P) is **unbounded** and (D) is **infeasible**.
3. (P) is **infeasible** and (D) is **unbounded**.
4. Both (P) and (D) are **feasible** (hence bounded) and their **optimal values coincide**: $\mathrm{OPT}(P) = \mathrm{OPT}(D)$.

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Case 4 is the main message: when both are feasible, the **duality gap is zero**.

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Important remarks:

- Duality does **not** apply to Integer Programming. It applies **only** to LP.

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Case 4 is the main message: when both are feasible, the **duality gap is zero**.

Important remarks:

- Duality does **not** apply to Integer Programming. It applies **only** to LP.
- The LP relaxations of Max Matching and Min VC are **always** dual to one another.
  - This has **no** bearing on their IP versions.
  - In particular, it does **not** imply that Min VC $\in$ P.

6.4 Strong Duality

Theorem 22 Strong Duality Theorem

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4. Both (P) and (D) are **feasible** (hence bounded) and their **optimal values coincide**: $\mathrm{OPT}(P) = \mathrm{OPT}(D)$.

Remark

Showing the min-VC LP is the dual of the max-matching LP follows directly from the recipe above.